

Path Tracking Control of Tracked Vehicles

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Abstract

A path tracking control algorithm for tracked surface drilling machines is presented. The general dynamic model of such vehicle is simulated including track-soil interactions. The controller is based on feed-back linearization of the equations with feedforward friction compensation. It linearizes the complex force-slip relationship to relate the states of the system to the inputs. The controller based on simplified models is applied to the detailed simulated model of the vehicle. The result is an accurate control of the lateral offset and forward velocity error along a path.

1 Introduction

The hazardous environment of a mine demands a high degree of autonomy of the mobile mining equipment to keep the human presence minimal. This research targets the surface drilling machines as used in open-pit mining, one of the most economical mining methods [9]. A surface drilling rig, equipped with two actuating tracks, drills long holes into the surface. The holes are subsequently filled with explosives. Drilling is a very slow process. Drilling in an area that has already experienced explosions, is considered a risk to the machine operator. Developing autonomous drills allows multi-unit supervision by a single remote operator, resulting in higher efficiency, and greater safety.

Despite recent advances in planning and control of autonomous mobile robots, there are many challenges in practical implementations. This work is a step towards converting real mining machines to autonomous units with on-board intelligence, operated by a remote user. To achieve this, a variety of problems have to be addressed, including modeling and control of tracked vehicles, navigation using GPS sensors, interactive user interface for teleoperation, and autonomous control in dynamic environments. The ongoing research at Ecole polytechnique aims at solv-

ing several of these issues within the SMART project (Sensori-Motor Augmented Reality for Telerobotics) and NSERC Chair for Mining Automation. In this paper, emphasis is given to the dynamics and path tracking control of tracked vehicles.

Skid steering is the most used steering mechanism for tracked vehicles. Despite its very simple manual usage, skid steering involves complicated ground-track interactions. It is still a subject of current research in ground mechanics. Traction forces are produced by the shear stress created by the track slippage relative to the ground. In the absence of an accurate model for this phenomena, experimental models relate the slippage and the shear stress [2, 1, 11, 4]. The resultant of the two track forces creates both the driving force needed for forward motion and the turning torque needed for steering. The main difficulty in controlling tracked vehicles is the high friction forces. Tracks with nonlinear and velocity- or force-dependent lateral friction terms represent a complicated dynamic non-holonomic problem for planning and control.

Numerous works address the modeling of the track forces such as [5] and [7], but few address the planning and autonomous control issues. The GPS-based navigation of surface drilling machines with no tracking control have been developed and are presently supplied by Aquila Company [8]. A planning method based on minimization of the lateral equality constraint violation is suggested in [10]. Estimates of track-soil interaction based on vehicle's kinematics and Kalman Filtering are simulated in [6].

Our approach uses the dynamic model of the vehicle for control and include the friction as additive nonlinear terms. By linearizing the force-slip relationship and compensating for the dynamic friction forces, we managed to simplify the controller design process. This approach allows for a path-tracking algorithm [3] which controls the lateral offset and forward velocity errors along a given path with a reasonable accuracy.

Starting from the basic equations of motion in Section 2, we formulate the control problem in in Sec-

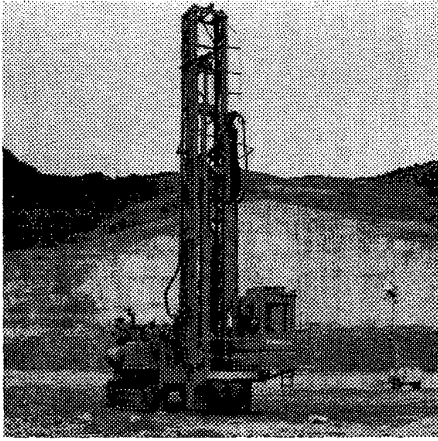


Figure 1: A surface drilling machine working in an open-pit mine.

tion 3. Results obtained from simulation are presented in Section 4. They are followed by conclusions in Section 5.

2 Dynamic Modeling

A vehicle has three degrees of freedom in the horizontal plane with the following assumptions. Velocity is small enough to neglect the centrifugal force on the turns. The weight of the vehicle is always equally divided on both tracks. X , Y , and θ represent the coordinates of the center of gravity in the fixed frame. A second coordinate system x - y is attached to the vehicle with its x axis along the vehicle as shown in Figure 2. In the latter F_i , R_i , F_o , and R_o are the traction and resistance forces exerted to the vehicle through the "inner" and "outer" tracks. Simplified equations of motion in x - y coordinates, provided by [6] and [10], are

$$\begin{aligned} M\ddot{x} &= (F_i + F_o - R_i - R_o) \\ M\ddot{y} &= 4f_y d_s \\ I\ddot{\psi} &= (F_o - R_o)b - (F_i - R_i)b - 2f_y(l^2 - d_s^2), \end{aligned} \quad (1)$$

where M is the mass of the vehicle, $d = \dot{y}/\dot{\psi}$, and $d_s = SAT(d, -l, l)$. The function $SAT(x, x_1, x_2) = \max\{x_1, \min\{x, x_2\}\}$ is the saturation function. Therefore, d_s is constrained to vary between $-l$ and

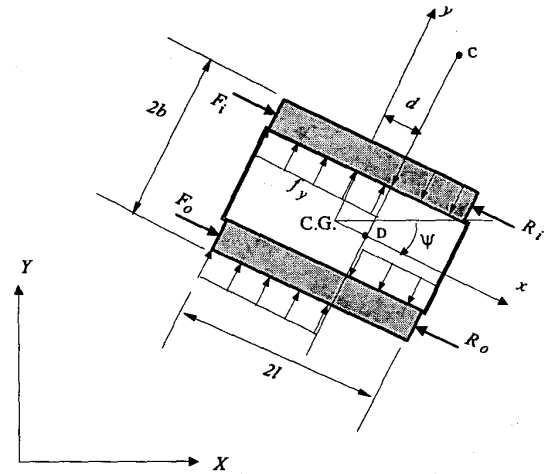


Figure 2: Tracked vehicle's force model.

l . These equations could also be written for the X - Y coordinate system using a planar rotation.

The force and friction terms of the right hand sides of the above equations have to be calculated. The point D located at a distance d from the center of gravity (C.G.) has a velocity along the x direction. Therefore, the instantaneous center of rotation, point C , is located on a line parallel to y axis passing through point D . The lateral velocity of any two points located on opposite sides D , are in the opposite directions. The lateral friction distribution is denoted by f_y . The direction of f_y at a point on the track depends on the direction of the speed of that point, while its magnitude is determined by the effective lateral friction coefficient μ_y and the normal force $Mg/2$. The track resistance force R depends on the longitudinal friction coefficient μ_x , and the normal force. The direction of R is determined by the track's speed with respect to the ground. These friction terms can be calculated by,

$$f_y = \frac{\mu_y Mg}{2l}, \quad \text{and} \quad R = \frac{\mu_x Mg}{2} \quad (2)$$

In this work, the ground is assumed to be a "dry clay," for which, $\mu_x = 0.2$ and given steel tracks, $\mu_y = 0.55$. In the particular case of Figure 2, the vehicle is turning in the positive direction, from X to Y .

The traction force F is related to the track slip i (see [6]) by

$$F = F_{max} \left[1 - \frac{\alpha}{|i|} \left(1 - e^{-|i|/\alpha} \right) \right] \cdot \text{sign}(i), \quad (3)$$

where $\alpha = k/2l$, and $k = 0.6 \text{ cm}$ for dry clay [11]. If A denotes the track area, the maximum tractive force $F_{max} = A\tau_{max}$, is limited by the maximum shear strength of the terrain $\tau_{max} = c + \tan(\phi)$. The constants are the apparent cohesion $c = 68.95 \text{ kpa}$, the normal pressure $p = Mg/2A$, and the angle of internal friction for dry clay $\phi = 0.5934$.

The inner and outer track slips are defined as

$$\begin{aligned} i_i &= \frac{V_i}{r\omega_i} & V_i &= \dot{x} - b\dot{\psi} - r\omega_i \\ i_o &= \frac{V_o}{r\omega_o}; & V_o &= \dot{x} + b\dot{\psi} - r\omega_o \end{aligned} \quad (4)$$

where r is the radius of the track roller, ω the angular velocity of the driving roller, and V is the slip velocity for each track, that is, the difference between the theoretical and actual velocity.

In the foregoing, friction models were developed only for a certain direction of motion, more general model is needed when the velocities are varying. The following example introduces a more generic way to present friction.

Assume a block with mass m on a surface with coefficient of friction μ , experiencing a horizontal input force F . The magnitude of the contact friction is $f = \mu \cdot m \cdot g$, leading to the equations of motion $m\ddot{x} = G(F, f, \dot{x})$. We define the function G , the “input-friction block,” as

$$G(F, f, \dot{x}) = \begin{cases} F - f\text{sign}(\dot{x}) & \dot{x} \neq 0 \\ 0 & \dot{x} = 0, |F| \leq f \\ F - f\text{sign}(F) & \dot{x} = 0, |F| > f \end{cases} \quad (5)$$

The vector G represents the resultant of the friction and the input force acting on the system. This notation provides a simple tool for expressing the multiple friction forces existing in the tracked vehicle dynamics.

A tracked vehicle is under three different types of friction; the lateral friction and the two track resistances. In this work, each of the friction forces is treated as one *input-friction block*. The traditional model used in soil mechanics (applying a total resistance $R = R_o + R_i$ against the traction as suggested in equations (1)). It cannot be used for all possible states. For example, if the vehicle is *turning in place* the direction of the resistance forces are against each other. Therefore, adding these forces will result in erroneous results in skid steering. As for which velocity should be used in determining the *input-friction block* for each track, we used the absolute velocity of the center of the track; $v_i = \dot{x} - b\dot{\psi}$ for the inner track, and $v_o = \dot{x} + b\dot{\psi}$ for the outer. Hence, for the forward direction, the resulting inner and outer

input-friction blocks are shown by $G_i = G(F_i, R_i, v_i)$, $G_o = G(F_o, R_o, v_o)$, calculated via equation (5).

The input-friction block can also be determined for the motion about the vertical axis, $G_\psi = G(M_F, M_r, \dot{\psi})$. The first argument represents the driving torque $M_F = b(G_o - G_i)$ produced by the track input-friction blocks in the forward direction. The second argument represents the resistance moment $M_r = 2f_y(l^2 - d_s^2)$, the integral of the lateral friction distribution. Finally, the equations of motion can be written as

$$\begin{aligned} M\ddot{x} &= G_i + G_o \\ M\ddot{y} &= -4f_y|d_s| \cdot \text{sign}(\dot{y}) \\ I\ddot{\psi} &= G_\psi \end{aligned} \quad (6)$$

Other models can be suggested and tested via simulations or experiments for systems undergoing several friction forces, studying which is beyond the scope of this paper.

3 Path Tracking Control

Control Problem

Friction forces in wheeled mobile robots are normally small enough to be neglected. For tracked vehicles, however, these forces are large and if neglected in the controller, produce large tracking errors. In this section, we introduce and verify a nonlinear control method for path tracking of tracked vehicles, keeping the accuracy at an acceptable level. The required accuracy for surface drilling operation is about 30 cm.

The overall control objective is to drive the vehicle from a starting point to any other point in a horizontal plane. This includes the “planning” and the “control” problems. As for the planning problem, we developed a software that generates a path by either using via points or by finding optimal path through the known obstacles. Figure 4 shows the integrated interface developed for planning and simulation of different types of vehicles.

Below, we focus only on the control problem with the assumption that the desired path is given. The desired path $\mathcal{P}$ is set of desired points $\mathbf{x}_d = (X_d, Y_d)$ in X - Y frame. A desired forward speed $\dot{x}_d$, is assigned to each point of the path $\mathcal{P}$ calculated by a simple velocity planner algorithm, details omitted. The lateral offset or error e_l , is defined as the distance between the actual position of the vehicle, $\mathbf{x} = (X, Y)$, and the closest point on $\mathcal{P}$, $e_l = \min\{|\mathbf{x}_d - \mathbf{x}|, \forall \mathbf{x}_d \in \mathcal{P}\}$. The velocity error is defined as $e_v = \dot{x}_d - \dot{x}$. The control

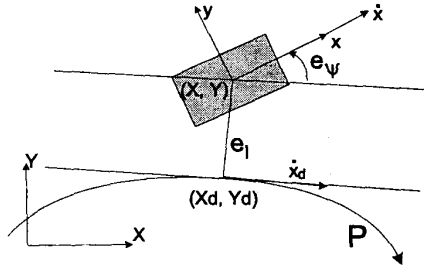


Figure 3: vehicle with offset from the desired path.

objective is to find the input $u \in \mathbb{R}^2$, $u = [\omega_i \ \omega_o]^T$, such that $\lim_{t \rightarrow \infty} e_l = 0$ and $\lim_{t \rightarrow \infty} e_\psi = 0$. However, some performance criteria could also be satisfied such as the response time. Figure 3 shows the vehicle with the lateral and heading offsets.

Controller Design

From the first of equations (6), the forward speed is only affected by the resultant of driving forces for both tracks, and from the last equation, the heading angle of the vehicle is only controlled by the difference between these track forces. For the system in hand, it is not possible to determine the input u to achieve a desired dynamics without any simplification. The exact feedback linearization is mathematically intractable. For an approximate solution, we split the problem into two. First, a simplified path tracking problem, find F_i and F_o as if they are the controlled inputs to the system. Second, find the corresponding ω_i and ω_o from the calculated forces of the first problem.

Part I: Any other kinematic property of the desired path required for control, such as the desired heading angle ψ_d , heading velocity $\dot{\psi}_d$, and the radius of curvature R_d , are calculated off-line at the planning time. The lateral offset from a path is

$$e_l = -(X - X_d) \sin(\psi_d) + (Y - Y_d) \cos(\psi_d), \quad (7)$$

see Figure 3. The following controller gives the reference heading rate

$$\Omega^* = \dot{\psi}_d - \frac{k_{r1}}{\dot{x}_d} e_l - k_{r2} e_\psi. \quad (8)$$

This computation is a mechanism to control the lateral offset. Here k_{r1} and k_{r2} are tunable rotation gains

and $e_\psi = \psi_d - \psi$ is the heading angle error. At this point the gains were tuned manually starting from a simple pole placement algorithm. The division of k_{r1} by $\dot{x}_d$ amplifies the corrections when the forward velocity is small and vice versa. Indeed, the gain could be divided by the real velocity instead of the desired one, except that it causes numerical problems at very small velocities. Pole placement for a second order dynamics of lateral error is used to provide an initial guess for the gains. They are tuned later. If the vehicle moves along the desired path, the last two terms become zero and Ω^* will be the same as $\dot{\psi}_d$, which implies the vehicle should stay on the path.

Next step is to make the vehicle to follow the reference heading rate Ω^* . Zero friction is assumed to derive the first part of the controller. Hence, for heading rate control, the difference of traction forces $F_\delta = F_{i,d} - F_{o,d}$ is to be found, the subscript “d” indicates that the forces are only the calculated desired ones, and not the real track forces. A simple controller

$$F_\delta = \frac{I}{b} k_\psi (\Omega^* - \dot{\psi}), \quad (9)$$

where k_ψ is a gain. Selection of such input is equivalent to creating a first order dynamics for the heading rate error $\dot{e}_\Omega + k_\psi e_\Omega = \Omega^*$, where $e_\Omega = \Omega^* - \dot{\psi}$. This follows by substituting equation (9) into equation (1), neglecting the friction terms. Gain selection is a simple pole placement for a first order system. Similarly, for the forward velocity controller a proportional integral action controller

$$F_\sigma = M \left[k_{t1} e_v + k_{t2} \int e_v dt \right], \quad (10)$$

yields the desired resultant force $F_\sigma = F_{o,d} + F_{i,d}$. Such controller results in the error equation $\dot{e}_v + k_{t1} e_v + k_{t2} \int e_v = \dot{x}_d$. The gains are selected by pole placement. The desired track forces are found using equations (9) and (10) as

$$F_{i,d} = \frac{F_\sigma - F_\delta}{2} \quad \text{and} \quad F_{o,d} = \frac{F_\sigma + F_\delta}{2}.$$

If the friction terms are included in the equations of motion, a simple feedforward compensation can be used to cancel their effect and reach at the same error equations. Only the dynamic friction terms can be compensated for, as they are fully determined by the states. In contrast, the value and direction of the static friction depends on input force and cannot be compensated by the input force itself. Finally, the foregoing yields the controller

$$F_{i,d} = \frac{F_\sigma - F_\delta}{2} - \frac{M_r * \text{sign}(\dot{\psi})}{2b} + R_i \text{sign}(v_i) \quad (11)$$

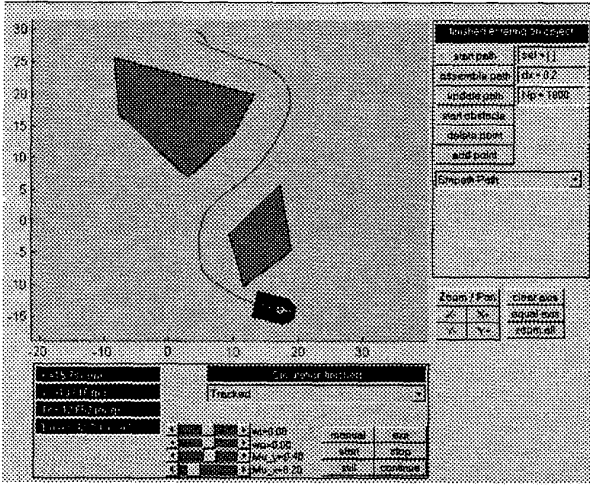


Figure 4: The planning and control interface.

$$F_{o,d} = \frac{F_{\sigma} + F_{\delta}}{2} + \frac{M_r * \text{sign}(\dot{\psi})}{2b} + R_o \text{sign}(v_o) \quad (12)$$

Part II: estimating the inputs ω_i and ω_o from desired forces $F_{i,d}$ and $F_{o,d}$ using equation (3). Inversion of equation (3) is not analytically possible; however, the expansion of the exponential term for small $|i|/\alpha$ solves this problem. If all the terms of higher order than two are neglected, equation (3) can be rewritten as

$$F = F_{max} \left[1 - \frac{\alpha}{|i|} \left(1 - \left(1 - \frac{|i|}{\alpha} + \frac{|i|^2}{2\alpha^2} \right) \right) \right] \text{sign}(i).$$

From $\text{sign}(i) = |i|/i$, the above equation simplifies to

$$F = \left(\frac{i}{2\alpha} \right) F_{max}$$

This relationship is linear and yields the desired slips

$$\begin{aligned} i_{i,d} &= 2\alpha \text{SAT}\left(\frac{F_{i,d}}{F_{max}}, -1, 1\right) \\ i_{o,d} &= 2\alpha \text{SAT}\left(\frac{F_{o,d}}{F_{max}}, -1, 1\right) \end{aligned} \quad (13)$$

This linearization is valid for the range of slips observed in the simulations. The inputs can now be found in terms of the desired slips

$$\omega_i = \frac{\dot{x} - b\dot{\psi}}{r(1 + i_{i,d})}, \text{ and } \omega_o = \frac{\dot{x} + b\dot{\psi}}{r(1 + i_{o,d})}. \quad (14)$$

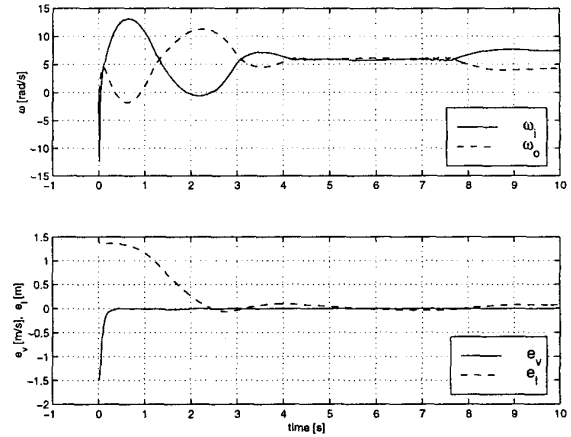


Figure 5: Top figure shows the track velocities and the bottom one shows how the lateral offset and speed errors converge to zero. A forward velocity of 1.5m/s was sought.

4 Results

The dynamic model of the track vehicle as described by equations (6) is simulated (see Figure 4 for screen shot). The controller described by equation (8) to (14) is implemented. The parameters are selected for a typical vehicle and the gains are designed using pole placement for the linearized dynamics of the system. Table 1 presents the numerical settings used for the gains and system parameters in the simulation runs. Figure 6 shows the actual traction forces and

$k_{r1} = 0.33$	$k_{r2} = 5$	$l = 2.5m$
$k_{t1} = 30$	$k_{t2} = 0.36$	$I = 40e3kgm^2$
$k_{\psi} = 4$	$r = 0.25m$	$b = 2.0m$

Table 1: Numerical settings for gains and parameters for a nominal surface drilling machine.

the corresponding track slips.

The system is always perturbed at the beginning of the run, from its desired path, to examine the convergence of the controller. Figure 5 depicts the required track velocities and how fast the tracking errors diminish for a typical path tracking experiment. Decomposition of the controller into separate loops for speed, angular rate, and the heading controllers gives the possibility of explicit calculation of the gains for further “tuning”. In addition, it provided a suitable means in designing a fast dynamics for the speed error e_v and slow dynamics for the lateral error e_l . The bottom graph of Figure 5 illustrates a fast convergence of

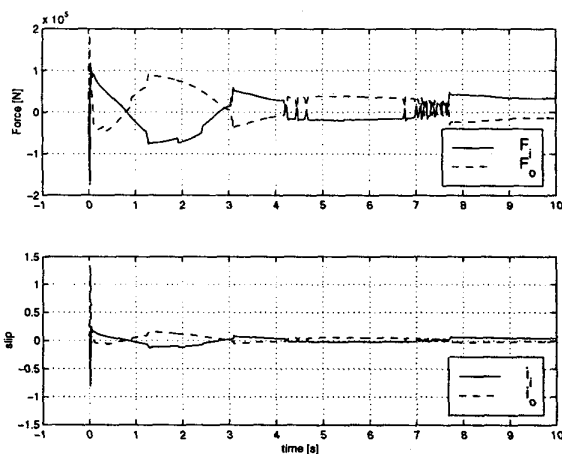


Figure 6: Top figure shows the traction forces and the bottom one shows the track slips.

$\dot{x} \rightarrow \dot{x}_d$ and a slow convergence of $e_l \rightarrow 0$. Fast convergence of the actual speed to the desired one also justifies the choice of using $\dot{x}_d$ instead of $\dot{x}$ in equation (8).

Finally, for the same path tracking controller, the effect of friction compensation is investigated. The upper panel in Figure 7 shows the lateral error for the case of no friction compensation, and the lower panel presents the same variable for when the compensation block is active.

5 Conclusions

A dynamic friction model was used to create a dynamic model for tracked vehicles. A path tracking controller was built based on a simplified friction model. The controller resulted in small steady state errors, despite the high friction forces of the tracked vehicles. Tracking errors of the controller are compared to those of a controller with no friction compensation. Future work, in addition to experimental implementation, will study the effects of the actuator dynamics, the engine-track dynamics, and estimate the critical parameters of the system. The vehicle state will be estimated using GPS sensors and plugged in to the proposed algorithm.

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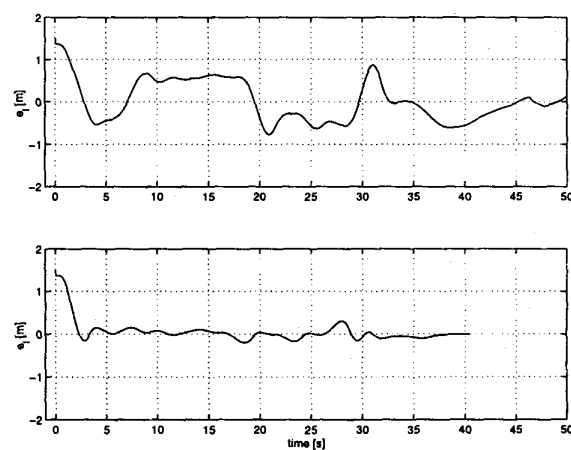


Figure 7: The lateral error. Top, for controller without friction compensation, and bottom with friction compensation.

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